Geometric Origin of the Charge Dimension in Discrete Gnomonic Projection: 5-Dimensional Spacetime and Charge Quantization

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DOI: 10.5281/zenodo.19435162

April 2026

Research Note (Geometric Investigation)

Abstract

In the previous paper [2], we showed that the isotropic geodesic equation on the discrete Minkowski lattice $\mathbb{Z}^{1+N}$ has no solutions for $N = 3$ (arithmetic anomaly) but admits solutions for $N = 4$. This paper identifies the fifth dimension with the charge (Coulomb) dimension and extends the gnomonic projection to $S^5(R)$.

No new coupling constants are introduced. The sole coupling constant C (dimension: length) is retained, and a conversion coefficient $Q = C/e_0$ (e_0: minimum charge quantum) is defined. The coordinate $\tilde{\phi} = q/e_0 \in \mathbb{Z}$ becomes an integer charge number, and charge quantization is derived from the discrete lattice without compactification. The $1/r^2$ Coulomb law is derived from the gnomonic mapping in the charge direction, and combined with Lorentz invariance from [2], Maxwell's equations in their entirety are geometrically derived. The difference in polarization structure between electromagnetic waves (spin-1) and gravitational waves (spin-2) emerges from the difference in observability of the curvature directions.

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1 Introduction: Confirmation of Discretization Premises

1.1 Premise: The Discrete Gnomonic Mapping Has Been Verified

This paper takes the **discrete gnomonic projection** established in [2] as its starting point. The following results have already been verified:

1. The coupling constant C discretizes coordinates: $\tilde{x}^\mu = x^\mu/C \in \mathbb{Z}$.
2. Under $w = C/R \in \mathbb{Z}_{\geq 0}$, the discrete gnomonic mapping $\tilde{Y}^\mu = \tilde{x}^\mu/\tilde{L}$ is defined.
3. All metrics are quadratic: $\tilde{\sigma}^2 = \sum_\mu \epsilon_\mu (\tilde{x}^\mu)^2$, $\tilde{L}^2 = 1 + w^2\tilde{\sigma}^2$.
4. Euclidean: $\tilde{L}^2 \geq 1 > 0$ at all lattice points (Theorem 6.1(i) of [2]).

1.2 Fundamental Reason for Quadratic Metrics

On the discrete lattice $\mathbb{Z}^n$, each coordinate takes positive and negative integers. A metric must be sign-invariant: $(-n)^2 = n^2$ for all $n \in \mathbb{Z}$. The quadratic form is the minimal

structure guaranteeing this:

$$\text{Discrete lattice} \implies \text{Negative coordinates} \implies \text{Sign invariance required} \implies (\tilde{x}^\mu)^2 \tag{1.1}$$

Remark (Physical meaning of negative coordinate values)**.** *For spatial axes: isotropy ($\pm n$ indistinguishable). For time: convention (arrow of time). For charge $\phi = Qq$: physical distinction (positive/negative charge). In all cases, the metric $(\tilde{x}^\mu)^2$ handles negative values; the physical distinction is preserved in the coordinate value, not the metric.*

1.3 Number-Theoretic Motivation: Why Five Dimensions

From Remark 6.2 of [2] and [3]: on $\mathbb{Z}^{1+N}$, the isotropic geodesic equation $l^2 + t^2 = Nx^2$ has solutions iff every prime $p \equiv 3 \pmod 4$ in N appears with even exponent (Fermat). $N = 3$ violates this ($3 \equiv 3$); $N = 4$ satisfies it ($4 = 2^2$). **Discrete isotropy requires at least $N = 4$ spatial dimensions.**

1.4 Observability of the Charge Dimension

In Axiom 3.1 of [2], unobservability of R led to imaginary rotation (Lorentzian signature). The charge dimension ϕ is **observable** (charge can be measured), so Axiom B is **not applied**: $\epsilon_4 = +1$.

Dimension	Observability	Axiom B applied?
Y^{n+1} (R-direction)	Unobservable	Yes $\to$ sign reversal
$\phi = Qq$ (charge)	**Observable**	**No** $\to$ sign unchanged

1.5 Organization

§2: 5D gnomonic mapping. §3: Coefficient Q and charge quantization. §4: 5D discrete mapping. §5: Recovery of discrete isotropy. §6: 5D Einstein tensor. §7: Consistency. §8: Future tasks and electromagnetic field.

2 Extension of the 5-Dimensional Gnomonic Mapping

2.1 Coordinate System

$$(x^1, x^2, x^3, x^4, x^5) = (x, y, z, \phi, ct) \tag{2.1}$$

Signature: $\epsilon_\mu = (+1, +1, +1, +1, -1)$.

2.2 5D Gnomonic Mapping

Applying [1] to $n = 5$:

$$\Phi : \Pi_R \to S^5(R), \quad Y^\mu = \frac{Rx^\mu}{l}, \quad Y^6 = \frac{R^2}{l} \tag{2.3}$$

$$l^2 = R^2 + x^2 + y^2 + z^2 + Q^2q^2 - c^2t^2 \tag{2.4}$$

Pullback metric:

$$g_{\mu\nu} = \frac{R^2}{l^2}\left(\epsilon_\mu \delta_{\mu\nu} - \frac{\epsilon_\mu \epsilon_\nu x_\mu x_\nu}{l^2}\right) \tag{2.5}$$

3 Conversion Coefficient Q and Charge Quantization

3.1 Structural Symmetry with c

	Time $\to$ Length	Charge $\to$ Length
Coefficient	c	Q
Formula	$x^5 = ct$	$x^4 = Qq$
Dimension	m/s	m/C

3.2 Determination of Q

Discretization $\tilde{\phi} = Qq/C \in \mathbb{Z}$ requires a minimum charge quantum e_0:

$$Q = \frac{C}{e_0}, \qquad \tilde{\phi} = \frac{q}{e_0} \in \mathbb{Z} \tag{3.3}$$

With $e_0 = e/3$ (quark charge):

$$Q = \frac{3C}{e} \approx 3.03 \times 10^{-16} \text{ m/C} \tag{3.5}$$

3.3 Charge Quantization

Theorem 3.1 (Charge quantization)**.** *Under $\tilde{\phi} \in \mathbb{Z}$ and $Q = C/e_0$:*

$$q = n \cdot e_0, \qquad n \in \mathbb{Z} \tag{3.7}$$

Remark (Compactification unnecessary)**.** *Unlike Kaluza–Klein theory [4, 5], charge quantization here derives from the discrete lattice $\tilde{\phi} \in \mathbb{Z}$, not from periodicity. The ϕ-direction is non-compact.*

Remark (Discrete coordinates for known particles)**.** *With $e_0 = e/3$: proton ($\tilde{\phi} = +3$), electron (-3), u quark ($+2$), d quark (-1), neutrino (0). All integers; $\tilde{\phi}^2$ identical for $\pm$ values.*

4 5-Dimensional Discrete Gnomonic Mapping

4.1 Discretization

$$\tilde{x} = x/C,\ \tilde{y} = y/C,\ \tilde{z} = z/C,\ \tilde{\phi} = q/e_0,\ \tilde{t} = ct/C\ \in \mathbb{Z} \tag{4.1}$$

4.2 Discrete Metric

$$\tilde{L}^2 = 1 + w^2\tilde{\sigma}^2, \qquad \tilde{\sigma}^2 = \tilde{x}^2 + \tilde{y}^2 + \tilde{z}^2 + \tilde{\phi}^2 - \tilde{t}^2 \tag{4.2}$$

4.3 Regularity (5D)

Theorem 4.1 (5D regularity)**.** ***(i)*** *Euclidean: $\tilde{L}^2 \geq 1 > 0$ globally.*

(ii) *Lorentzian: horizon at $c^2t^2 = R^2 + x^2 + y^2 + z^2 + Q^2q^2$. The charge term $Q^2q^2 \geq 0$* ***expands*** *the regular region compared to 4D.*

5 Recovery of Discrete Isotropy at $N = 4$

For $N = 4$: $4 = 2^2$ contains no prime $p \equiv 3 \pmod 4$. The equation $l^2 + t^2 = 4x^2$ has solutions. Minimal: $l = 0, t = 2, x = 1$.

Proposition 5.1 (Recovery of discrete isotropy)**.** *No isotropic null geodesic exists on $\mathbb{Z}^{1+3}$, but they exist on $\mathbb{Z}^{1+4}$ with the charge dimension. The charge dimension is number-theoretically required for discrete isotropy.*

6 5-Dimensional Einstein Tensor

Applying Proposition 2.1 of [1] to $n = 5$:

$$\Lambda_5 = \frac{(5-1)(5-2)}{2R^2} = \frac{6}{R^2}, \qquad G_{\mu\nu} + \frac{6}{R^2} g_{\mu\nu} = 0 \tag{6.1–6.2}$$

Compared to 4D: $\Lambda_5 = 2\Lambda_4$.

7 Consistency Checks

Coupling constants: Only C. $Q = C/e_0$ is a conversion coefficient, not a free parameter.

Quantity	Conversion	Discrete coord.	Quantization
Position	x [m]	$\tilde{x} = x/C \in \mathbb{Z}$	Length
Time	t [s] $\to ct$ [m]	$\tilde{t} = ct/C \in \mathbb{Z}$	Time
Charge	q [C] $\to Qq$ [m]	$\tilde{\phi} = q/e_0 \in \mathbb{Z}$	**Charge**

Proposition 7.1 (Safety of negative coordinates)**.** *In the discrete gnomonic mapping, $\tilde{L}^2$ is invariant under sign change of any $\tilde{x}^\mu$, since $(\tilde{x}^\mu)^2 = (-\tilde{x}^\mu)^2$.*

8 Future Tasks and the Electromagnetic Field

8.1 Derivation of the Fine-Structure Constant

$\alpha = e^2/(4\pi\epsilon_0\hbar c) \approx 1/137$ is not derived in this paper. Connection to Wyler's formula [6] via the $SO(5,1)$ symmetry group of $S^5(R)$ is an important future task.

8.2 Electromagnetic Field from the Charge Direction

8.2.1 Directed Curvature in the ϕ-Direction

	Gravity	Electromagnetism
Curvature direction	R (unobservable)	$\phi = Qq$ (**observable, directed**)
Curvature radius	R	$\phi_0 = Qq_0$ (source charge)
Nature	Uniform	**Directed** (sign of q_0)

8.2.2 4D Beltrami Metric ($R = \infty$)

Source charge q_0 curves (x, y, z, t)-space with radius $\phi_0 = Qq_0$:

$$g_{\mu\nu} = \frac{\phi_0^2}{l_q^2}\left(\eta_{\mu\nu} - \frac{x_\mu x_\nu}{l_q^2}\right), \qquad l_q^2 = \phi_0^2 + \sigma^2 \tag{8.1}$$

where $\eta_{\mu\nu} = \mathrm{diag}(+1, +1, +1, -1)$, $\sigma^2 = x^2 + y^2 + z^2 - c^2t^2$.

8.2.3 Inverse Metric and Christoffel Symbols

With $\Sigma = l_q^2/\phi_0^2$, $K = 1/\phi_0^2$:

$$g^{\mu\nu} = \Sigma(\eta^{\mu\nu} + Kx^\mu x^\nu), \qquad \Gamma^\rho_{\mu\nu} = -\frac{K}{\Sigma}(x_\mu\delta^\rho_\nu + x_\nu\delta^\rho_\mu) \tag{8.2–8.3}$$

8.2.4 Derivation of the $1/r^2$ Law

The gnomonic embedding gives $Y^i = \phi_0 x^i/l_q$. For a spatially static particle ($x^i = r_i =$ const), time progression changes l_q. The second time derivative at $t = 0$:

$$\left.\frac{d^2Y^i}{dt^2}\right|_{t=0} = \frac{\phi_0\, r_i\, c^2}{(\phi_0^2 + r^2)^{3/2}} \tag{8.7}$$

In the limit $r \gg |\phi_0|$:

$$\boxed{\frac{d^2Y^i}{dt^2} \approx \frac{Qq_0 \cdot c^2}{r^2}\,\hat{r}_i} \tag{8.8}$$

This is the $1/r^2$ Coulomb law.

8.2.5 Test Particle Response

Remark (Test charge response is geometric)**.** *A neutral particle ($q_1 = 0$) sits at the tangent point $\phi_1 = 0$ where the metric is flat; no force. A charged particle ($q_1 \neq 0$) is displaced from the tangent point; curvature effects are proportional to $|q_1|$:*

$$a \sim \frac{Q^2 q_0 q_1 c^2}{r^2} \tag{8.9}$$

Remark (Absence of the equivalence principle)**.** *For gravity: R is unobservable, so all particles respond identically (equivalence principle). For EM: ϕ is observable, so response depends on q_1 (no equivalence principle). The equivalence principle is a* ***consequence*** *of the unobservability of R.*

8.2.6 Polarity and Directionality

The sign of $\phi_0 = Qq_0$ determines field direction: $q_0 > 0 \to$ outward; $q_0 < 0 \to$ inward. Gravity (R undirected) is always attractive; EM (ϕ directed) can be attractive or repulsive.

8.2.7 Polarization: Spin-1 vs Spin-2

Proposition 8.1 (Polarization and observability)**.** *EM waves are spin-1 (vector polarization) because the ϕ-direction is directed. Gravitational waves are spin-2 (tensor polarization) because the R-direction is undirected.*

	EM waves	Gravitational waves
Curvature dir.	ϕ (directed)	R (undirected)
Perturbation	Vector (selects direction)	Symmetric tensor
Spin	**1**	**2**

8.2.8 Complete Form of Maxwell's Equations

Proposition 8.2 (Geometric derivation of Maxwell's equations)**.** *Coulomb's law (8.8) from this paper, combined with Lorentz invariance from [2], yields all four Maxwell equations. The magnetic field* $\mathbf{B}$ *is the Lorentz transform of* $\mathbf{E}$*; Faraday's and Ampère's laws follow from covariance.*

Maxwell's equation	Derivation
$\nabla \cdot \mathbf{E} = \rho/\epsilon_0$	Eq. (8.8)
$\nabla \cdot \mathbf{B} = 0$	ϕ is 1D $\to$ no monopole d.o.f.
$\nabla \times \mathbf{E} = -\partial \mathbf{B}/\partial t$	Lorentz invariance [2]
$\nabla \times \mathbf{B} = \mu_0 \mathbf{J} + \mu_0 \epsilon_0 \partial \mathbf{E}/\partial t$	Lorentz invariance [2]

Summary of Results

Item	Result	Eq.	Status
Number-theoretic motivation	No solution at $N=3$; exists at $N=4$	§1.3	Fermat [3]
Charge observability	ϕ observable $\to \epsilon_4 = +1$	§1.4	Axioms A,B
Conversion coeff. Q	$Q = C/e_0$	(3.3)	Definition
Charge quantization	$q = ne_0$	(3.7)	Thm. 3.1
5D regularity	$\tilde{L}^2 \geq 1$; horizon expansion	(4.4)	Thm. 4.1
Discrete isotropy	$t = 2x$ at $N=4$	(5.2)	Prop. 5.1
5D Λ	$\Lambda_5 = 6/R^2$	(6.1)	Prop. 2.1
Coulomb's law	$d^2Y^i/dt^2 \approx Qq_0c^2/r^2$	(8.8)	§8.2
Equivalence principle	R unobs. $\to$ yes; ϕ obs. $\to$ no	Rem. 8.3	§8.2
Polarization	EM: spin-1; GW: spin-2	Prop. 8.1	§8.2
Maxwell's equations	Coulomb + Lorentz $\to$ complete	Prop. 8.2	§8.2

References

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This paper is confined to pure geometry and number theory. Electromagnetic dynamics, the fine-structure constant, and identification with existing theories are left for future research. Last updated: 2026-04-06